

Homological flat bands in strongly coupled $SU(N)$ Hamiltonian lattice gauge theory

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Abstract

We study the Kogut–Susskind Hamiltonian for pure $SU(N)$ gauge theory on a periodic cubic spatial lattice in the strong-coupling regime. Projection to the fundamental, charge-odd one-plaquette manifold produces a three-orbital Bloch Hamiltonian. We show that its second-order hopping is not an engineered flat-band model: non-Abelian group integration generates the oriented edge–face boundary Gram operator $B(\mathbf{k})B(\mathbf{k})^\dagger$, where $B = \partial_2^\dagger$. The exact all-rank coefficient is

$$t_N = \frac{2N(N^2 - 4)}{(N^2 - 1)(2N^2 - 1)(4N^2 - 9)} > 0, \quad N \geq 3.$$

Consequently the projected spectrum contains one momentum-independent level and a doubly degenerate branch shifted by $t_N u^2 q(\mathbf{k})$, with $q = 4 \sum_j \sin^2(k_j/2)$. The flat carrier is $\ker \partial_2$: elementary cube boundaries give compact localized states, while three wrapping sheets complete the carrier on the torus. For $L \geq 3$, its exact multiplicity is $(L^3 - 1) + 3 = L^3 + 2$. For $SU(3)$, exact third-order perturbation theory remains in the same boundary-factorized algebra and gives

$$H_{\text{eff},-} = E_{\text{flat}}(u)I + \left(\frac{5}{612}u^2 + \frac{1975}{124848}u^3 \right) BB^\dagger + O(u^4),$$

with $E_{\text{flat}} = 8/3 + u + 11u^2/306 - 109151u^3/249696$. The result is a finite-volume, finite-order theorem for a projected flux sector; it is not a continuum glueball or mass-gap theorem.

1 Introduction

Hamiltonian strong-coupling expansions organize lattice gauge dynamics around the electric vacuum and its flux excitations [1, 2, 3]. In three spatial dimensions, a fundamental plaquette excitation carries one of three face orientations. At second order, neighboring faces communicate through a shared link, so the projected problem resembles a three-orbital tight-binding model. The signs of those matrix elements, however, are fixed by the orientation of the plaquette boundary and by charge conjugation. The resulting interference is the central object of this paper.

Flat bands generated by incidence or Gram matrices have a long history. Local topology, line-graph constructions, compact localized states, and the need for noncontractible states on periodic systems are well established [4, 5, 6, 7, 8]. Rectangular factorizations and index arguments provide a general language for such zero modes [9]. Closely related curl-free or closed-surface structures occur in two-form spin liquids [10], and a recent independent construction treats face-graph and higher-cell flat bands directly [12]. We therefore do not claim that incidence flatness, cellular compact localized states, or homological counting are new in isolation.

The result here is instead dynamical. Starting from non-Abelian $SU(N)$ Hamiltonian lattice gauge theory, exact strong-coupling matrix elements generate an *oriented edge–face* Gram operator in the charge-odd one-plaquette sector. Group representation theory fixes its coefficient, while the cubical chain complex fixes its kernel and spectrum. The separation is exact:

$$\text{color dynamics} \longrightarrow t_N, \quad \text{cellular geometry} \longrightarrow BB^\dagger.$$

This produces four main statements.

1. For every $N \geq 3$, the complete momentum dependence at order u^2 is $t_N u^2 BB^\dagger$, with the positive rational function t_N displayed in the abstract.
2. The Bloch spectrum of $BB^\dagger$ is $\{0, q(\mathbf{k}), q(\mathbf{k})\}$. The zero branch is the space of boundary-free face amplitudes.
3. On the periodic L^3 cubical complex, the zero-mode count is $L^3 + 2$: $L^3 - 1$ independent cube boundaries and three harmonic wrapping sheets.
4. For $SU(3)$, all third-order processes capable of leaving the shared-link incidence algebra vanish. The scalar and hopping coefficients can therefore be assembled exactly, proving rigidity of the flat branch through $O(u^3)$.

These statements concern the effective Hamiltonian projected to the degenerate one-plaquette manifold. Virtual multi-plaquette states are retained in the resolvents, but the external spectral problem is not the full gauge-theory Hilbert space. At $\mathbf{k} = 0$ the three incidence

branches touch; the finite-volume separation collapses as L^{-2} . Nothing below proves convergence of the strong-coupling series, persistence under unrestricted higher-order operators, identification with the lightest physical glueball, or a continuum mass gap.

2 Hamiltonian and projected sector

Let $\Lambda_L = (\mathbb{Z}/L\mathbb{Z})^3$ be a cubic spatial lattice with periodic boundary conditions. In units of the electric coupling, the Kogut–Susskind Hamiltonian is

$$H_\beta = \frac{1}{2} \sum_\ell C_2(\ell) + \beta_N \sum_p \left(1 - \frac{1}{N} \text{Re Tr } U_p \right). \quad (1)$$

After dropping the additive constant, write

$$H = H_E - uV, \quad V = \sum_p (\chi_p + \bar{\chi}_p), \quad u = \frac{\beta_N}{2N}, \quad (2)$$

where $\chi_p = \text{Tr } U_p$ is the fundamental character. For $SU(3)$, $u = \beta_3/6 = 1/g_H^4$ in the convention used below. The definition $u = \beta_N/(2N)$, rather than the $SU(3)$ -specific final equality, is the all-rank coupling convention.

The electric vacuum $|0\rangle$ has the trivial representation on every link. For $N \geq 3$, normalized one-plaquette states of definite charge conjugation may be chosen as

$$|p, \pm\rangle = \frac{\chi_p \pm \bar{\chi}_p}{\sqrt{2}} |0\rangle, \quad (3)$$

up to an immaterial phase in the odd combination. Character orthogonality gives unit norm. Each of the four boundary links carries the fundamental or antifundamental representation, so the unperturbed energy is

$$E_{F,N} = 4 \left(\frac{1}{2} C_F \right) = 2C_F = \frac{N^2 - 1}{N}, \quad C_F = \frac{N^2 - 1}{2N}. \quad (4)$$

For $SU(2)$, complex conjugation is an inner gauge transformation and the charge-odd state vanishes; this is why the construction begins at $N = 3$.

Let P_- project onto the charge-odd one-plaquette manifold and $Q = 1 - P_-$. Degenerate perturbation theory gives a formal effective operator

$$H_{\text{eff},-} = P_- H_E P_- + uH_1 + u^2 H_2 + u^3 H_3 + \cdots, \quad (5)$$

where every H_r contains the appropriate Q -space resolvents. Thus projection fixes the external states but does not discard virtual multi-plaquette sectors. Translation invariance reduces the external problem at each $\mathbf{k}$ to three face orbitals, which we order as (12), (13), (23).

3 Oriented incidence and the exact carrier

The periodic cubical complex has the cellular chain sequence

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \xrightarrow{\partial_1} C_0, \quad \partial_2 \partial_3 = 0. \quad (6)$$

The one-plaquette amplitudes are elements of C_2 . Their signed amplitude leaking into links is $\partial_2 c$. This is a cellular closed-surface condition, not the ordinary microscopic vertex Gauss law: the Wilson-loop basis already satisfies the latter through $\partial_1 \partial_2 = 0$.

For

$$d_j = e^{ik_j} - 1, \quad q(\mathbf{k}) = \sum_{j=1}^3 |d_j|^2 = 4 \sum_{j=1}^3 \sin^2 \frac{k_j}{2}, \quad (7)$$

choose the face-to-link Bloch incidence matrix

$$B(\mathbf{k}) = \partial_2(\mathbf{k})^\dagger = \begin{pmatrix} d_2 & -d_1 & 0 \\ d_3 & 0 & -d_1 \\ 0 & d_3 & -d_2 \end{pmatrix}. \quad (8)$$

Orbital gauge changes multiply rows or columns by phases and do not alter the following statements.

Theorem 1 (Incidence factorization). *Let*

$$\mathbf{w}(\mathbf{k}) = (\bar{d}_3 \quad -\bar{d}_2 \quad \bar{d}_1)^T. \quad (9)$$

Then

$$BB^\dagger = qI - \mathbf{w}\mathbf{w}^\dagger, \quad B^\dagger \mathbf{w} = 0, \quad (10)$$

and therefore

$$\text{spec}(BB^\dagger) = \{0, q, q\}. \quad (11)$$

If $S = BB^\dagger - 4I$ is the signed shared-link adjacency, then

$$\text{spec } S(\mathbf{k}) = \{-4, q(\mathbf{k}) - 4, q(\mathbf{k}) - 4\}. \quad (12)$$

Proof. Multiplying Equation (8) by its adjoint gives the first identity in Equation (10) entry by entry. Since $\|\mathbf{w}\|^2 = q$, the rank-one operator $qI - \mathbf{w}\mathbf{w}^\dagger$ annihilates $\mathbf{w}$ and acts as qI on its orthogonal complement. Subtracting $4I$ gives Equation (12). $\square$

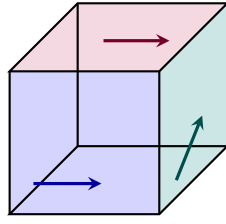
For $\mathbf{k} \neq 0$, the carrier is one dimensional and is represented by $\mathbf{w}(\mathbf{k})$. At Γ , $B = 0$, so all three branches meet. The normalized Bloch eigenvector has no direction-independent limit at Γ ; this is the familiar singular-flat-band mechanism behind the incompleteness of a purely compact localized basis [6, 7].

Theorem 2 (Finite-torus carrier). *For the periodic cubic cellulation of T^3 ,*

$$\ker \partial_2 = \text{im } \partial_3 \oplus \mathcal{H}_2, \quad (13)$$

with

$$\dim \text{im } \partial_3 = L^3 - 1, \quad \dim \mathcal{H}_2 = 3. \quad (14)$$



cube boundary $\partial_3 c$
six oriented faces

$$C_3 \xrightarrow{\partial_3} C_2 \xrightarrow{\partial_2} C_1 \rightarrow \ker \partial_2 = \text{im } \partial_3 \oplus \mathcal{H}_2$$

$$\partial_2 \partial_3 = 0 \quad (L^3 - 1) \text{ cube boundaries} \\ \text{local carrier} \quad + 3 \text{ wrapping sheets}$$

Figure 1: The cellular origin of the carrier. An oriented elementary cube boundary is a compact localized state because every edge occurs twice with opposite induced signs. On a torus, cube boundaries span $\text{im } \partial_3$, but three harmonic wrapping sheets are also required.

Hence

$$\boxed{\dim \ker \partial_2 = L^3 + 2.} \quad (15)$$

Elementary cube boundaries span $\text{im } \partial_3$; three independent wrapping sheets span $\mathcal{H}_2 \simeq H_2(T^3)$.

Proof. There are L^3 three-cells and no four-cells. Since $b_3(T^3) = 1$, rank-nullity gives $\text{rank } \partial_3 = L^3 - 1$. Discrete Hodge decomposition [11] gives $Z_2 = \ker \partial_2 = B_2 \oplus \mathcal{H}_2$, and $\dim \mathcal{H}_2 = b_2(T^3) = 3$. Adding the dimensions proves Equation (15). The single relation among translated elementary cube boundaries is the oriented sum over all cubes. $\square$

The number $L^3 + 2$ is the nullity of the down boundary Laplacian, not a Betti number. The Betti contribution is only the final 3. For the ordinary twelve-neighbor Bloch convention we assume $L \geq 3$; at $L = 2$, coincident periodic neighbors require an explicit multigraph convention. The first nonzero incidence eigenvalue is

$$q_{\min} = 4 \sin^2 \frac{\pi}{L}, \quad (16)$$

which will control the coefficient-level finite-volume separation.

4 Exact all-rank dynamics at second order

We now derive the coefficient multiplying the geometrical operator. Two plaquettes sharing a link leave six non-shared fundamental links after the external contractions. On the shared link, the possible fusion channels depend on the relative induced orientation:

$$F \otimes \bar{F} = \mathbf{1} \oplus \text{Adj}, \quad F \otimes F = \Lambda^2 F \oplus \text{Sym}^2 F. \quad (17)$$

Let R be one of these irreducible channels. Isotropy of the normalized shared-link tensor assigns squared norm d_R/N^2 to the channel projector. The six nonshared half-links contribute $3C_F$ and the fused link contributes $C_R/2$, whereas the external one-plaquette energy is $2C_F$. The resolvent weight is therefore

$$w_R = -\frac{d_R/N^2}{C_F + C_R/2}. \quad (18)$$

The required dimensions and Casimirs are

$$\begin{aligned} d_1 = 1, \quad C_1 = 0, \quad d_{\text{Adj}} = N^2 - 1, \quad C_{\text{Adj}} = N, \\ d_{\Lambda^2 F} = \frac{N(N-1)}{2}, \quad C_{\Lambda^2 F} = \frac{(N+1)(N-2)}{N}, \\ d_{\text{Sym}^2 F} = \frac{N(N+1)}{2}, \quad C_{\text{Sym}^2 F} = \frac{(N-1)(N+2)}{N}. \end{aligned} \quad (19)$$

Substitution into Equation (18) and summation within the two orientation families gives

$$\mathcal{A}_N = -\frac{2N^3}{(N^2-1)(2N^2-1)}, \quad (20)$$

$$\mathcal{B}_N = -\frac{4N(N^2-2)}{(N^2-1)(4N^2-9)}. \quad (21)$$

The charge-odd combination takes their signed difference.

Theorem 3 (All-rank shared-link law). *For every integer $N \geq 3$, the charge-odd one-plaquette effective Hamiltonian has the form*

$$H_{\text{eff},-}^{(N)}(\mathbf{k}, u) = E_{\text{flat},N}(u)I + t_N u^2 B(\mathbf{k})B(\mathbf{k})^\dagger + O(u^3), \quad (22)$$

where the scalar $E_{\text{flat},N}$ is momentum independent and

$$\boxed{t_N = \mathcal{B}_N - \mathcal{A}_N = \frac{2N(N^2-4)}{(N^2-1)(2N^2-1)(4N^2-9)} > 0.} \quad (23)$$

The three bands through second order are

$$E_{\text{flat},N}(u), \quad E_{\text{flat},N}(u) + t_N u^2 q(\mathbf{k}) \quad (\text{twice}). \quad (24)$$

Proof. The diagonal and within-plaquette terms are scalars in face-orbital space. Every second-order inter-plaquette process uses one shared link, and its relative orientation is precisely the incidence product appearing in $S = BB^\dagger - 4I$. The contributions in Equations (20) and (21) therefore assemble as $d_N I + t_N S$. Absorbing $-4t_N$ into the scalar gives Equation (22). The factorization theorem then gives Equation (24). For $N \geq 3$, every denominator in Equation (23) is positive and $N^2 - 4 > 0$, proving strict positivity. $\square$

Two useful consequences are

$$t_3 = \frac{5}{612}, \quad t_N = \frac{1}{4N^3} - \frac{1}{16N^5} - \frac{77}{64N^7} + O(N^{-9}). \quad (25)$$

The width of the *full retained charge-odd manifold* is

$$W_N^{(-)} = 12t_N u^2 + O(u^3) \sim \frac{3u^2}{N^3}. \quad (26)$$

This N^{-3} statement concerns the stiffness of the dispersive doublet and the full-manifold width. It is not motion of the lowest branch, which remains flat at this order.

For comparison, removing the orientation signs gives an unsigned incidence matrix $N(\mathbf{k})$ with $A + 4I = NN^\dagger$. Its adjacency spectra at Γ and R are $\{12, 0, 0\}$ and $\{-4, -4, -4\}$, respectively. No momentum-independent eigenvalue can belong to both sets. The flat branch is therefore not a generic consequence of three face orbitals or cubic symmetry; the charge-odd orientation signs are essential.

5 $SU(3)$ rigidity through third order

We now specialize to $SU(3)$, where the full scalar ledger through third order is available in exact arithmetic. The within-plaquette charge-odd tower contributes

$$\frac{8}{3} + u + \frac{1}{2}u^2 + \frac{7}{32}u^3. \quad (27)$$

At second order, shared-link self-energy and hopping terms reduce the flat coefficient to

$$\frac{1}{2} + 12 \left(-\frac{11}{306} \right) - 4 \left(\frac{5}{612} \right) = \frac{11}{306}, \quad (28)$$

when organized in the scalar-plus- $BB^\dagger$ convention. Equivalently, the charge-odd diagonal and signed-hop coefficients before the shift are $7/102$ and $5/612$.

At third order, processes confined to one plaquette or to a two-plaquette domino renormalize the same scalar and shared-link structures. A new connected geometry could involve three distinct plaquettes. The following lemma is the dynamical reason no new Bloch shape appears at this order.

Lemma 4 (Third-order lifter vanishing). *Every connected three-distinct-plaquette numerator that could map the charge-odd one-plaquette manifold outside the scalar-plus-shared-link incidence algebra vanishes. Geometrically, each candidate leaves at least one fundamental boundary link unmatched, so Haar integration over that link is zero.*

Remark 5. The lemma is deliberately scoped to the new three-distinct-plaquette lifters. Repeated-plaquette and domino processes do not vanish; they produce the nonzero scalar and hopping coefficients below. Exact enumeration groups the candidate lifters into 32 five-trace numerator classes and returns zero for every class.

The nonzero third-order per-neighbor quantities are

$$b_3 = \frac{1975}{124848}, \quad \text{leak}_3 = -\frac{12331}{249696}. \quad (29)$$

Combining them with the tower coefficient yields

$$d_3 = \frac{7}{32} + 12\text{leak}_3 - 4b_3 = -\frac{109151}{249696}. \quad (30)$$

Theorem 6 ($SU(3)$ third-order factorization). *In the charge-odd one-plaquette sector of $SU(3)$,*

$$\boxed{H_{\text{eff},-}(\mathbf{k}, u) = E_{\text{flat}}(u)I + \tau(u)B(\mathbf{k})B(\mathbf{k})^\dagger + O(u^4)}, \quad (31)$$

where

$$E_{\text{flat}}(u) = \frac{8}{3} + u + \frac{11}{306}u^2 - \frac{109151}{249696}u^3, \quad (32)$$

$$\tau(u) = \frac{5}{612}u^2 + \frac{1975}{124848}u^3. \quad (33)$$

Thus the complete three-band spectrum through $O(u^3)$ is

$$E_{\text{flat}}(u), \quad E_{\text{flat}}(u) + \tau(u)q(\mathbf{k}) \quad (\text{twice}). \quad (34)$$

Proof. The second-order result is Theorem 3 at $N = 3$. At third order, Lemma 4 removes every new connected three-plaquette lifter. The remaining linked contributions are the scalar leakage and signed shared-link hop in Equation (29); their assembly is Equation (30). Hence the entire momentum dependence remains proportional to $BB^\dagger$, and Theorem 1 gives the spectrum. $\square$

The finite-volume coefficient-level gap between the carrier and the nearest dispersive state is therefore

$$\Delta_L(u) = 4\tau(u)\sin^2 \frac{\pi}{L} + O(u^4). \quad (35)$$

For sufficiently small positive u , the displayed coefficient is positive. Since it scales as L^{-2} , it does not provide volume-uniform spectral isolation.

Hamer's independent 1989 Hamiltonian series for the 1^{+-} channel [3] supplies a non-load-bearing cross-check of the rest-frame coefficients. With his $x = 2u$ convention and the Hamiltonian normalization $H_{\text{here}} = W/2$, the converted coefficients through x^3 agree with Equation (32) to the precision printed in that paper; the cubic comparison differs by approximately 1.1×10^{-12} . The analytic assignment of the Γ -point orbital triplet to T_1^{+-} is consistent with that comparison. We nevertheless reserve the term *glueball mass* for a full-sector spectral statement, which is not proved here.

6 What the homology protects

The lower-order result extends immediately to any correction that remains inside the boundary-factorized operator class.

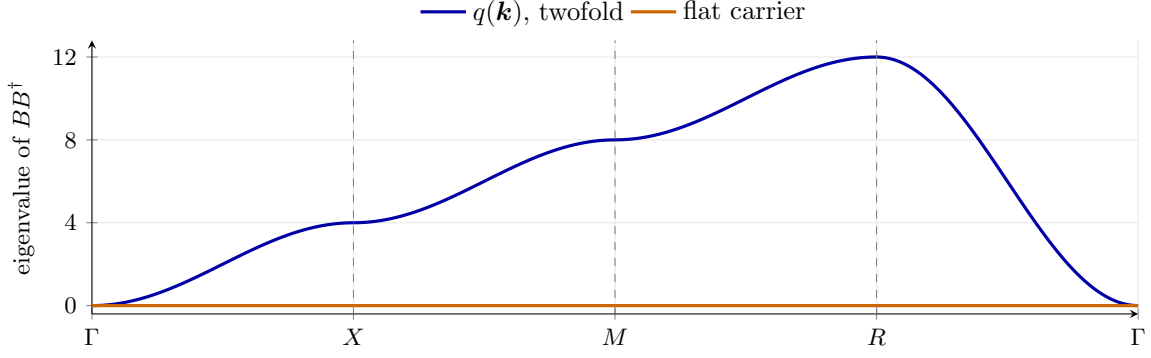


Figure 2: Universal incidence spectrum along $\Gamma-X-M-R-\Gamma$. Color dynamics only rescales the blue doublet by $t_N u^2$, or by $\tau(u)$ for the $SU(3)$ third-order result. All three branches touch at Γ .

Proposition 7 (Boundary-factorized rigidity). *Suppose an order- r Bloch correction has the form*

$$H_r(\mathbf{k}) = a_r I + b_r S(\mathbf{k}) + B(\mathbf{k}) M_r(\mathbf{k}) B(\mathbf{k})^\dagger. \quad (36)$$

Then for every $\mathbf{k} \neq 0$,

$$H_r(\mathbf{k}) \mathbf{w}(\mathbf{k}) = (a_r - 4b_r) \mathbf{w}(\mathbf{k}). \quad (37)$$

Thus this correction shifts the carrier without dispersing it.

Proof. Use $B^\dagger \mathbf{w} = 0$ and $S\mathbf{w} = -4\mathbf{w}$. $\square$

In real space the same statement is the orthogonality of the two cellular Laplacians

$$L_2^\downarrow = \partial_2^\dagger \partial_2, \quad L_2^\uparrow = \partial_3 \partial_3^\dagger, \quad L_2^\downarrow L_2^\uparrow = L_2^\uparrow L_2^\downarrow = 0. \quad (38)$$

Every polynomial in $L_2^\downarrow$ and $L_2^\uparrow$ acts on the harmonic subspace by its constant term. A term proportional to $L_2^\uparrow$, however, disperses the cube-boundary component while leaving the three harmonic sheets pinned. Homology therefore distinguishes local boundaries from true harmonic cycles.

The proposition is not an unconditional all-orders theorem. The set of operators $BMB^\dagger$ is not a two-sided ideal in $\text{End}(C_2)$, and a general higher-order gauge-theory correction need not factor through links. Cubic symmetry alone permits carrier-projected fourth-order shapes built from

$$\begin{aligned} &1, \quad q, \quad e_2, \quad \frac{4e_2}{q}, \quad \frac{e_3}{q}, \\ e_2 &= \sum_{i < j} a_i a_j, \quad e_3 = a_1 a_2 a_3, \end{aligned} \quad (39)$$

where $a_i = 4 \sin^2(k_i/2)$. The rational shapes encode directional limits at Γ and need not belong to the boundary-factorized class.

Two exact fourth-order constructions in the underlying computation agree on the axial datum and on the appearance of mobility but disagree on one off-axis mixed-gradient coefficient. The target-blind linked-cluster calculation required to adjudicate that coefficient has not

completed. Accordingly, no fourth-order band, bandwidth, or derived higher-order quantity is used in this paper. This is a scientific boundary, not merely a choice of presentation.

7 Relation to prior flat-band constructions

It is useful to state precisely what is shared with, and what differs from, nearby constructions.

Line-graph and molecular-orbital models engineer a rectangular matrix R and obtain zero modes from $R^\dagger R$ or $RR^\dagger$ [5, 8, 9]. Here the same algebraic mechanism appears, but $R = \partial_2$ is selected by the orientation signs of charge-odd non-Abelian flux, and its prefactor is the exact representation-theoretic number t_N . The hopping ratio is therefore derived from the gauge Hamiltonian rather than tuned as a lattice design.

The two-form spin construction of Chung and Gingras [10] also features plaquette variables, curl constraints, flat interaction bands, and winding sectors. Its microscopic degrees of freedom and emergent $U(1)$ description differ from the projected $SU(N)$ flux operator studied here. Hazra’s contemporary face-graph construction [12] gives compact states and higher-cell generalizations using a designed face-incidence Hamiltonian. Our work was developed independently, and our new content relative to that framework is the non-Abelian strong-coupling derivation, the all-rank color coefficient, and the order-resolved $SU(3)$ rigidity.

The torus count likewise has a standard topological interpretation. Compact cube boundaries provide $L^3 - 1$ states, while singularity of the Bloch eigenvector forces noncontractible supplements [6, 7]. In the cubical gauge setting those supplements are exactly the three generators of $H_2(T^3)$.

8 Scope and physical interpretation

At Γ , the three face-orbital states transform as an axial vector of the cubic group. Together with parity even and charge conjugation odd, this gives the retained-space label T_1^{+-} . This label is an operator-space assignment. A physical glueball statement would require, at minimum, a volume-uniform dressed-operator overlap theorem, survival of the level after coupling to the complement of the one-plaquette external manifold, and a controlled continuum limit. None is implied by Theorems 3 and 6.

Several limitations should therefore remain attached to every use of the result:

- The strong-coupling series is treated as a formal finite-order expansion; convergence is not asserted.
- The carrier is macroscopically degenerate at finite L , but its nearest incidence separation vanishes as L^{-2} .
- Third-order rigidity is currently established for $SU(3)$, whereas the all-rank theorem stops at second order.
- The ordinary vertex Gauss law is already satisfied by the Wilson-loop basis. The additional equation $\partial_2 c = 0$ is a boundary-free two-chain condition.
- No disputed fourth-order coefficient, historical fifth-order series, continuum extrapolation, or mass-ratio prediction enters the theorem.

Within those boundaries, the result is exact and useful: it supplies a controlled operator seed whose geometry, degeneracy, rank scaling, and first three $SU(3)$ coefficients are known.

9 Reproducibility and evidence boundary

The analytic proof in this paper is self-contained. A portable standard-library Python program distributed with the source, `verify_core.py`, independently checks the rational coefficient ledger, the incidence identity at deterministic generic momenta, the chain condition $\partial_2 \partial_3 = 0$, and the finite-torus ranks for $L = 1, \dots, 4$.

The broader research repository is pinned at commit `ca3d440a7f93c17569e12d0511847505b6b72c5a`. At that commit, the recorded continuous-integration run reports 119 exact-rational checks, 29 numerical cross-checks, 349 repository tests, and 28 Lean theorems with no omitted proofs. Only a subset bears directly on the present paper. The relevant commands are

```
uv sync --frozen --all-extras
make verify
make check
make lean
```

and the paper-level checks can be reproduced with

```
python3 verify_core.py
```

Machine checks certify algebra relative to stated definitions and inputs; they do not replace the perturbative derivation, and they do not certify an uncompleted fourth-order linked-cluster construction. This distinction is why the theorem ends at $O(u^3)$.

10 Conclusions

Strong-coupling $SU(N)$ gauge dynamics and cubical topology separate cleanly in the charge-odd one-plaquette sector. The non-Abelian shared-link integrals produce the positive all-rank coefficient t_N , while the oriented boundary map produces an exact Gram operator. Its kernel is a flat band of boundary-free two-chains, with cube-boundary compact states and three harmonic wrapping sheets. For $SU(3)$, the same incidence algebra survives through third order, yielding the complete exact three-band spectrum in Equation (34).

The most important open step is equally sharp: a target-blind fourth-order contraction must decide the first off-axis carrier operator. Until that is done, higher-order bandwidth and continuum claims remain outside the theorem. The lower-order result stands independently as a dynamically generated homological flat band with exact color dependence.

A Channel weights in closed form

For completeness, substituting Equation (19) into Equation (18) gives

$$\begin{aligned} w_1 &= -\frac{2}{N(N-1)(N+1)}, \\ w_{\text{Adj}} &= -\frac{2(N-1)(N+1)}{N(2N^2-1)}, \\ w_{\Lambda^2 F} &= -\frac{N-1}{(N+1)(2N-3)}, \\ w_{\text{Sym}^2 F} &= -\frac{N+1}{(N-1)(2N+3)}. \end{aligned} \quad (40)$$

The mixed and like sums are

$$\mathcal{A}_N = w_1 + w_{\text{Adj}}, \quad \mathcal{B}_N = w_{\Lambda^2 F} + w_{\text{Sym}^2 F},$$

which simplify to Equations (20) and (21). Their difference has numerator $2N(N^2-4)$, proving both the $SU(2)$ zero and positivity for $N \geq 3$. Expanding in $z = 1/N^2$ gives

$$t_N = \frac{1}{4N^3} + \frac{1}{N^3} \left(-\frac{z}{16} - \frac{77z^2}{64} - \frac{1021z^3}{256} + \dots \right).$$

Table 1: Evidence map for the claims used in this paper. The term exact means rational or symbolic arithmetic; it does not mean an all-orders or continuum theorem.

Claim	Scope	Independent check
$BB^\dagger = qI - \mathbf{w}\mathbf{w}^\dagger$ and $\text{spec}(BB^\dagger) = \{0, q, q\}$ $\dim \ker \partial_2 = L^3 + 2$	all $\mathbf{k}$ periodic cubical complex	Direct multiplication; deterministic generic-momentum residuals in <code>verify_core.py</code> Homology proof; exact modular ranks for $L = 1, \dots, 4$
$t_N = \mathcal{B}_N - \mathcal{A}_N > 0$	integer $N \geq 3$	Channel-projector derivation; exact rational anchors; denominator-cleared Lean identities in the repository
$SU(3)$ scalar and hopping through u^3	projected charge-odd sector	Exact rational ledger; independent comparison with Hamer’s printed rest-frame series
Third-order incidence rigidity	$SU(3)$	Bare-link proof; exact enumeration of 32 lifter classes

B Finite-volume chain calculation

The cubical boundary of an oriented (i, j) face based at x , with $i < j$, is

$$\partial_2[x; i, j] = [x + \mathbf{e}_i; j] - [x; j] - [x + \mathbf{e}_j; i] + [x; i]. \quad (41)$$

The boundary of a cube is

$$\begin{aligned} \partial_3[x; 1, 2, 3] &= [x + \mathbf{e}_1; 2, 3] - [x; 2, 3] \\ &\quad - [x + \mathbf{e}_2; 1, 3] + [x; 1, 3] \\ &\quad + [x + \mathbf{e}_3; 1, 2] - [x; 1, 2]. \end{aligned} \quad (42)$$

Substitution of Equation (42) into Equation (41) cancels every edge pairwise. On the torus, $\ker \partial_3$ is generated by the sum of all oriented cubes, so $\text{rank } \partial_3 = L^3 - 1$. Since $H_2(T^3) \simeq \mathbb{R}^3$, the count in Theorem 2 follows.

C Exact $SU(3)$ ledger

The coefficient assembly used in Theorem 6 is summarized in Table 2. All entries are rational.

Table 2: $SU(3)$ charge-odd coefficient ledger.

Order	Quantity	Value
u^0	electric plaquette	8/3
u^1	scalar	1
u^2	$BB^\dagger$ coefficient	5/612
u^2	flat scalar	11/306
u^3	$BB^\dagger$ coefficient	1975/124848
u^3	per-neighbor leakage	-12331/249696
u^3	flat scalar	-109151/249696

The last line is the exact identity

$$-\frac{109151}{249696} = \frac{7}{32} + 12 \left(-\frac{12331}{249696} \right) - 4 \left(\frac{1975}{124848} \right).$$

D Data and source availability

The manuscript source and the portable core verifier accompany this version. The complete research repository

is identified by the immutable commit hash given in Section 9; it contains the theory graph, provenance ledger, exact checks, and formal algebra. The unresolved fourth-order production calculation is explicitly excluded from the results reported here.

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